

Code Update – July 5, 2026

Today I began developing the firmware for the ESP32 telemetry system using C++ and PlatformIO. I created the initial software architecture by organizing the project into separate header (.h) and source (.cpp) files to improve code organization and scalability. I implemented a TelemetryData structure to store battery voltage, motor temperature, RPM, vehicle speed, pitch, and roll information. I also created functions to generate simulated telemetry data and display it through the serial monitor. Although the telemetry values are currently simulated, this software framework will allow real sensor data to be integrated with minimal changes as the hardware is added. Finally, I integrated the firmware project into the GitHub repository to begin tracking firmware development through version control.